

Tianrun Hu 胡天润

Ph.D. Student · School of Computing, National University of Singapore

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Research Interests

My research focuses on real-world mobile manipulation, with an emphasis on *reasoning in unknown and dynamic environments*. I develop representations and planning methods that enable robots to reason under partial knowledge, direct sensing toward task-relevant uncertainty, and adapt their actions as the environment changes.

Education

National University of Singapore, Ph.D. in Computer Science Aug 2026 – Present
School of Computing · Advised by Prof. Harold Soh · A*STAR Graduate Scholar (Computing)

Nanyang Technological University, B.Eng. in Computer Engineering, Distinction Honours Jul 2020 – Jul 2024
College of Computing and Data Science · GPA 4.47/5.0

Research & Professional Experience

Robot Navigation for Humanitarian Assistance and Disaster Relief Jul 2026 – Present
Stanford University · Research collaboration with Dr. Yiqing Xu under Prof. Jiajun Wu

Research Engineer, Full-Time Aug 2024 – Jul 2026
Adaptive Computing Lab, Smart Systems Institute (SSI), NUS · Advised by Prof. David Hsu

- Led visibility-aware mobile grasping, jointly planning gaze and base motion in unknown, dynamic environments.
- Built an imitation learning system for multiple robots and pip-installable motion planners on OMPL and VAMP.

Research Assistant, Part-Time · LinS Lab, NUS · Advised by Dr. Lin Shao Sep 2023 – Jun 2024

- Built a Gaussian Splatting-to-mesh indoor reconstruction pipeline and generated data for ManiFoundation (IROS 2024).

Machine Learning Engineer, Internship · Continental AG, Singapore Jan 2023 – Jun 2023

- Developed core SLAM modules for an autonomous forklift, focusing on robust localization and mapping in industrial sites.

Selected Publications

- [1] Agent Priors-guided Policy Learning In Submission
Puming Jiang*, **Tianrun Hu***, Haozhe Du, Yibo Li, Zhiwei Xue, Xinhui Li, Harold Soh [arXiv] [website] [code]
- [2] Visibility-Aware Mobile Grasping in Dynamic Environments RSS 2026 Workshop (ExWBC), Oral Presentation
Tianrun Hu*, Anxing Xiao, David Hsu, Hanbo Zhang [arXiv] [website] [demo] [code]
- [3] Hypothesis-driven Model Expansion under Uncertainty for Open-World Robot Planning RSS 2026
Anxing Xiao*, Hanbo Zhang, **Tianrun Hu**, David Hsu [arXiv] [website] [video] [code]
- [4] Chain-of-Action: Trajectory Autoregressive Modeling for Robotic Manipulation NeurIPS 2025
Wenbo Zhang*, **Tianrun Hu**, Yanyuan Qiao, Hanbo Zhang, Yuchu Qin, Yang Li, Jiajun Liu, Tao Kong, Lingqiao Liu, Xiao Ma [arXiv] [website] [code]
- [5] MimicFunc: Imitating Tool Manipulation from a Single Human Video via Functional Correspondence CoRL 2025
Chao Tang*, Anxing Xiao, Yuhong Deng, **Tianrun Hu**, Wenlong Dong, Hanbo Zhang, David Hsu, Hong Zhang [arXiv] [website] [code]
- [6] Robi Butler: Remote Multimodal Interactions with Household Robot Assistant ICRA 2025
Anxing Xiao*, Nuwan Janaka, **Tianrun Hu**, Anshul Gupta, Kaixin Li, Cunjun Yu, David Hsu [arXiv] [website] [video]

Projects

Motion Planning Libraries [Autolife] [Fetch] [Wheelchair] [Bimanual Franka] [Whole-Body VAMP]
CPU motion planning in milliseconds on OMPL and VAMP, with one pip-installable stack per robot and a shared Python API. The Fetch and Wheelchair stacks add FLASK kinodynamic planning.

Uniform Framework for Data Gathering and Behavior Cloning [code]
One imitation-learning framework across teleoperation interfaces and robots, from demonstrations to training ACT and diffusion policies.

Digital Twin and 3D Reconstruction [docs] [GS toolkit] [digital twin]
Gaussian Splatting tools for indoor 3D mesh reconstruction, and a lab digital twin built on them for CS6244 and mobile manipulation research.

Service & Teaching

Reviewer CoRL 2026, IEEE RA-L 2025 Workshop Human-Centric Mobile Manipulation, RSS 2026 (helped organize)

Teaching CS6244 Advanced Topics in Robotics, NUS, 2025 Open source Contributor to OMPL

Honors & Awards

A*STAR Graduate Scholarship (Computing), A*STAR Graduate Academy 2026

Dean's List (Top 5%), Nanyang Technological University 2024

Science & Engineering Undergraduate (SM2) Scholarship, NTU 2019 – 2024

Skills

Software Python, C, C++, Java · ROS 1 & ROS 2, PyTorch, OMPL, VAMP, CasADi, OpenCV, Isaac Sim, Gazebo

Hardware Motors and sensors, Arduino, basic mechanical design Languages Chinese (native), English (fluent)